

Sudip Sharma, Ph.D.

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RESEARCH PROFILE

Computational biologist developing scalable statistical and machine learning frameworks for phylogenomics and genomic inferences. My work integrates evolutionary biology with statistics and machine learning to uncover genotype–phenotype relationships, identify unreliable phylogenetic signals, and improve inference from large-scale genomic data. I am particularly interested in applying these approaches to problems in comparative genomics, evolutionary medicine, cancer evolution, functional genomics, and viral evolution.

RESEARCH INTERESTS

Evolutionary genomics; phylogenomics; machine learning for genotype–phenotype association; statistical methods for large-scale genomic inference; comparative genomics; molecular evolution; viral evolution; functional genomics; genomic prediction.

EDUCATION

Temple University, Philadelphia, PA Fall 2018 – Fall 2023

Ph.D. in Biology

Advisor: Dr. Sudhir Kumar

University of Dhaka, Dhaka, Bangladesh 2016 – 2017

M.Sc. in Applied Statistics

University of Dhaka, Dhaka, Bangladesh 2012 – 2015

B.Sc. in Applied Statistics

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow Fall 2023 – Present

Institute for Genomics and Evolutionary Medicine, Temple University

Mentor: Dr. Sudhir Kumar

Research Assistant Summer 2019 – Summer 2023

Institute for Genomics and Evolutionary Medicine, Temple University

Mentor: Dr. Sudhir Kumar

Teaching Assistant Fall 2018 – Spring 2021

Department of Biology, Temple University

Statistician February 2018 – July 2018

International Centre for Diarrhoeal Disease Research, Bangladesh

TECHNICAL SKILLS

Programming: Python, R, C++, Bash, Windows batch scripting, HPC/Slurm

Phylogenetics and Genomics: IQ-TREE, MEGA, ASTRAL, sequence alignment analysis, phylogenomics

Machine Learning and Statistics: Sparse learning, Regression models, Causal inference, Bootstrap methods, Genomic Risk Score modeling

R/Python Tools: Biostrings, ape, phangorn, phytools, data.table, pROC, PLINK, gwaslab

SELECTED SCIENTIFIC CONTRIBUTIONS

- Co-developed the Little Bootstrap framework for scalable phylogenomic inference, published in *Nature Computational Science*.
- Co-developed ModelTamer, a subsampling-based approach for efficient substitution model selection in phylogenomics.
- Co-developed DrPhylo based on evolutionary sparse learning to identify fragile clades and genomic features associated with phylogenetic hypotheses.
- Contributed to MyESL, a software platform for applying evolutionary sparse learning to molecular phylogenetics and genomics.
- Contributed to methods and software for large-scale molecular evolutionary analysis, including MEGA.

PUBLICATIONS

Primary Author Publications

1. Sharma, S. and Kumar, S. Fast and accurate bootstrap confidence limits on genome-scale phylogenies using little bootstraps. *Nature Computational Science* **1**, 573–577 (2021).
2. Sharma, S. and Kumar, S. Taming the selection of optimal substitution models in phylogenomics. *Molecular Biology and Evolution* **39** (2022).
3. Sharma, S. *Big Data Phylogenomics: Methods and Applications*. Ph.D. Dissertation, Temple University (2023).
4. Sharma, S. and Kumar, S. Discovering fragile clades and causal sequences in phylogenomics by evolutionary sparse learning. *Molecular Biology and Evolution* **41**, msae131 (2024).
5. Sanderford, M., Sharma, S., Stecher, G., Suleski, M., Liu, J., Ye, J. and Kumar, S. MyESL: a software for evolutionary sparse learning in molecular phylogenetics and genomics. *Molecular Biology and Evolution* **42**, msaf224 (2025).
6. Sharma, S. and Kumar, S. Robust and efficient confidence limits for phylogenomic inference of organismal relationships. *Molecular Biology and Evolution* **42**, msaf296 (2025).
7. Sharma, S., Sanderford, M., and Kumar, S. MyESL-v2: Scalable and Efficient Sparse Learning for Genomic Data. *In Preperation* (2026).

Collaborative Publications

8. Allard, J. B., Sharma, S., Patel, R., Sanderford, M., Tamura, K., Vucetic, S., Gerhard, G. S. and Kumar, S. Evolutionary sparse learning reveals the shared genetic basis of convergent traits. *Nature Communications* **16**, 3217 (2025).
9. Tao, Q., Sharma, S., Tamura, K. and Kumar, S. R3F: An R package for evolutionary dates, rates, and priors using the relative rate framework. *arXiv* (2025).

10. Kumar, S., Stecher, G., Suleski, M., Sanderford, M., Sharma, S. and Tamura, K. MEGA12: Molecular Evolutionary Genetic Analysis version 12 for adaptive and green computing. *Molecular Biology and Evolution* **41**, msae263 (2024).
11. Craig, J. M., Gerhard, G. S., Sharma, S., Yankovskiy, A., Miura, S. and Kumar, S. Methods for estimating personal disease risk and phylogenetic diversity of hematopoietic stem cells. *Molecular Biology and Evolution* **41**, msad279 (2024).
12. Barba-Montoya, J., Sharma, S. and Kumar, S. Molecular timetrees using relaxed clocks and uncertain phylogenies. *Frontiers in Bioinformatics* (2023).
13. Kumar, S., Tao, Q., Weaver, S., Sanderford, M., Caraballo-Ortiz, M. A., Sharma, S., Pond, S. L. and Miura, S. An evolutionary portrait of the progenitor SARS-CoV-2 and its dominant offshoots in the COVID-19 pandemic. *Molecular Biology and Evolution* **38**, 3046–3059 (2021).
14. Kumar, S. and Sharma, S. Evolutionary sparse learning for phylogenomics. *Molecular Biology and Evolution* **38**, 4674–4682 (2021).
15. Shchubelka, K., ..., Sharma, S., ..., Kumar, S.,..., Oleksynk, T. K. A collaborative genomics platform for T1D in Eastern and Central Europe. *Metabologia* (2026).
16. Kumar, S., Tamura, K., and Sharma, S. Phylogenomic subsampling and upsampling for efficient evolutionary analyses of big data. *Molecular Biology and Evolution* (2026).

SOFTWARE AND COMPUTATIONAL CONTRIBUTIONS

- **MyESL:** Software platform for evolutionary sparse learning in molecular phylogenetics and genomics.
- **Phylogenomic subsampling and upsampling pipelines:** Computational workflows for scalable bootstrap inference, model selection, and parameter estimation.
- **MEGA-related contributions:** Contributions to molecular evolutionary analysis workflows and large-scale phylogenetic applications.

GRANTS AND AWARDS

Travel Grant , Bangladesh Sweden Trust Fund	December 2018
Supported travel to the United States for doctoral studies.	
Merit Scholarship , ISRT, University of Dhaka	January 2016 – May 2017
Merit-based scholarship for graduate students with the highest grades.	
Merit Scholarship , ISRT, University of Dhaka	January 2013 – May 2015
Merit-based scholarship for undergraduate students with the highest grades.	

CONFERENCE PRESENTATIONS

Evolution Meeting 2026 , Cleveland, Ohio	June 2026
Talk: Net Bootstrap: Robust and Efficient Confidence Limits for Phylogenomics	
Evolution Meeting 2025 , Athens, Georgia	June 2025
Talk: Discovering Fragile Clades and Causal Sequences in Phylogenomics using Evolutionary Sparse Learning	
Evolution Meeting 2023 , Albuquerque, New Mexico	June 2023
Poster: MEGA 12 for faster, greener, and accurate big data analytics on personal to powerful computers	

SMBE 2022 , Virtual	July 2022
Talk: Taming the selection of optimal substitution models in phylogenomics	
Evolution Meeting 2022 , Cleveland, Ohio	June 2022
Talk: Fast and accurate bootstrap confidence limits on genome-scale maximum likelihood phylogenies using little bootstraps	
SMBE 2021 , Virtual	July 2021
Talk: Fast and accurate bootstrap confidence limits on genome-scale phylogenies using little bootstraps	

PROFESSIONAL SERVICE

Journal Reviewer

- Molecular Biology and Evolution
- Genome Biology and Evolution
- Bioinformatics Advances
- Frontiers in Bioinformatics
- Genome Biology
- Scientific Reports

PROFESSIONAL MEMBERSHIPS

Society of Molecular Biology and Evolution	2021 – Present
Society for the Study of Evolution	2019 – Present